

CLEF '25 notes and interesting posters

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Contents

1 Introductions

1.1 Keynote: Sameer Antani, AI for Medicine

- Data in medicine is difficult, often biased (i.e. more prevalence of disease vs. natural distribution due to only imaging correct skin cancer)
- AI in medicine must be multimodal (e.g. there is always an order attached to an image!)
- Synthesis as a remedy:
 - Clinical training for people
 - Fill data gaps/sparse data,
 - Problems: Hallucinations, not rule-based (anatomy, diseases, ...)
- Evaluation of synthetic data: what is the specific impact of it being added?
 - Generalization? or just improvements?
 - Hallucinations eval?

Note: CLEF has changed reviews to focus on methodology instead of raw numbers (was good for our submission I guess?)

1.2 Conference Sessions I (Best of CLEF 2024)

1.2.1 Humour Classification According to Genre and Technique by Fine-tuning LLMs

- Add the definitions of the classes into prompts
- Tree-based LM classifier

1.2.2 Language-based Mixture of Transformers for Sexism Identification in Social Networks

- Use ensemble of domain-specific models (models trained on Twitter, same source domain!)
- Model mixture: variation (either half-half, 75 percent or only dominant)
- Some fine-tuning
- Q: how are they mixed? i.e. at what stage, dynamically chosen? based on what??

1.2.3 Robustness of Misinformation Classification Systems to Adversarial Examples Through BeamAttack

- Counterfactuals for classification, which minimal modification has to be done to change output (CheckTHAT task)
- Effectively: how good is the adversarial attack? (Similarity - Levensthein, Effectiveness scored)
- BERT-Attack:
 - Which word: based on word importance (using logits, each word is masked after each other - calculate probability)
 - What to insert: masked AE (e.g. RoBERTa)

- DeepWordBug: replace characters/typos
- Theirs: use beam-search for improved search of replacement
- Tree width + depth of search are hyperparameters
- Disadvantage: needs a lot of evaluation whether they affect the classifier

1.3 Labs Overview

394 Papers, Labs: 13 old + 1 new

1.3.1 Overview of LifeCLEF 2025: Challenges on Species Presence Prediction and Identification, and Individual Animal Identification

- @Simon? iNaturalist :)
- For environmental monitoring
 - BirdCLEF: Sound classification (3k participants - 50k price pool!)
 - PlantCLEF: detection of plants in plots of land
 - GeoCLEF: Multimodal Classification for species
 - AnimalCLEF: Open-Set classification (New! individuals)
 - FungiCLEF: Few-shot classification, with multi-modal description
- Paper count not correlated to price pools ;)
- Foundational models were the winners
- Compared to humans: only experts can outperform these models, have strong location prior

1.3.2 Overview of BioASQ 13

- 6 tasks, 6 languages, 3 doc types
- 17 participant in GutBrainIE

1.3.3 Overview of Touché 2025: Argumentation Systems

- Debate simulation,
 - Evaluation Grice’s maxims of cooperation
 - Systems often switched sides or admitted defeat!
- analysis
 - ParlaMint: multilingual debates, scores on english best
- image arguments (generation+analysis); eval → core aspects of images are evaluated; best submission extracted aspects and prompted image gen
- Advertisement in RAG: Generate (eval: classifier), and detect ads in responses (AdBlock for LMs)
<https://touche.webis.de/clef25/touche25-web/advertisement-detection.html#task> (eval: yes/no)

1.3.4 Overview of the CLEF 2025 JOKER Lab: Humour in Machine

- LMs not able to deal with humor etc.
- humor-aware IR
 - Search for jokes on topics
 - Manual + LM generated jokes, mixed with non-humor (wikipedia)
 - Eval: humor + traditional IR metrics; way better results this year!
- Translate puns
 - Wordplay consistent accross EN-FR translation
 - Q: is the annotation for the “funny word” given to the participants?
 - Eval: consistent meaning of translations, location based of the wordplay
- Onomastic Wordplay Translation

- e.g. often in Harry Potter, Asterix, ...
- Used in training sets
- EN-FR
- Q: copyright, could GPT have been trained on the source material?

1.3.5 LongEval at CLEF 2025: Longitudinal Evaluation of IR Systems on Web and Scientific Data

- training on evolving information needs over 9 months
- Trending queries and qrels (click models)
- On the TU Wien Research Dataset!

2 LifeCLEF 2025

2.1 Learning from Visual Data in the Wild (Oisín Mac Aodha)

- Growth in Biodiversity data ← iNaturalist,
- Range Maps of Species
 - downside of these citizen scientist approaches: spatially sparse - biased towards human locations, species distribution mismatched with iNaturalist observation
 - Very few expert Range maps...
 - LM generated range maps: only squares, very bad in relation to correct Range maps (interesting research topic?)
 - Idea: Sparse input of observation, output of range maps?
 - Presence detection - based on spatial embeddings $\odot$ species embeddings; need to be compact - fit on phones, improve offline CV species prediction (actually improves it! & is deployed on iNaturalist)
 - Spatial embeddings + species embeddings helps share data between low-observations and high-observation species
 - No absence data, only present data...
 - Visualization of high-dim vector on spatial data: PCA to 3D to RGB
 - Add text to context: as few as 5-10 observations from text works as text quite well
 - Joint training with representation learning for satellite images: Dense Retrieval of Text, Segmentation, ...

2.2 GeoLifeCLEF Overview

- Absence/Presence data, climate data, time series (climate)
- Very biased observations, people go places
- Test data: not only in-distribution, but OOD with new regions (with presence only)

Participant: Gleb Tikhonov

- Combination of a lot of handcrafted features and encoding systems
- Embedding of images, ...
- Averaging, cycling,...

2.2.1 PlantCLEF Overview

- Earlier: monospecies
- Now: multispecies, in singular images
 - Multiscale, variety of seasons,
 - Train: single plants, monospecies
 - Some non-annotated quadrats

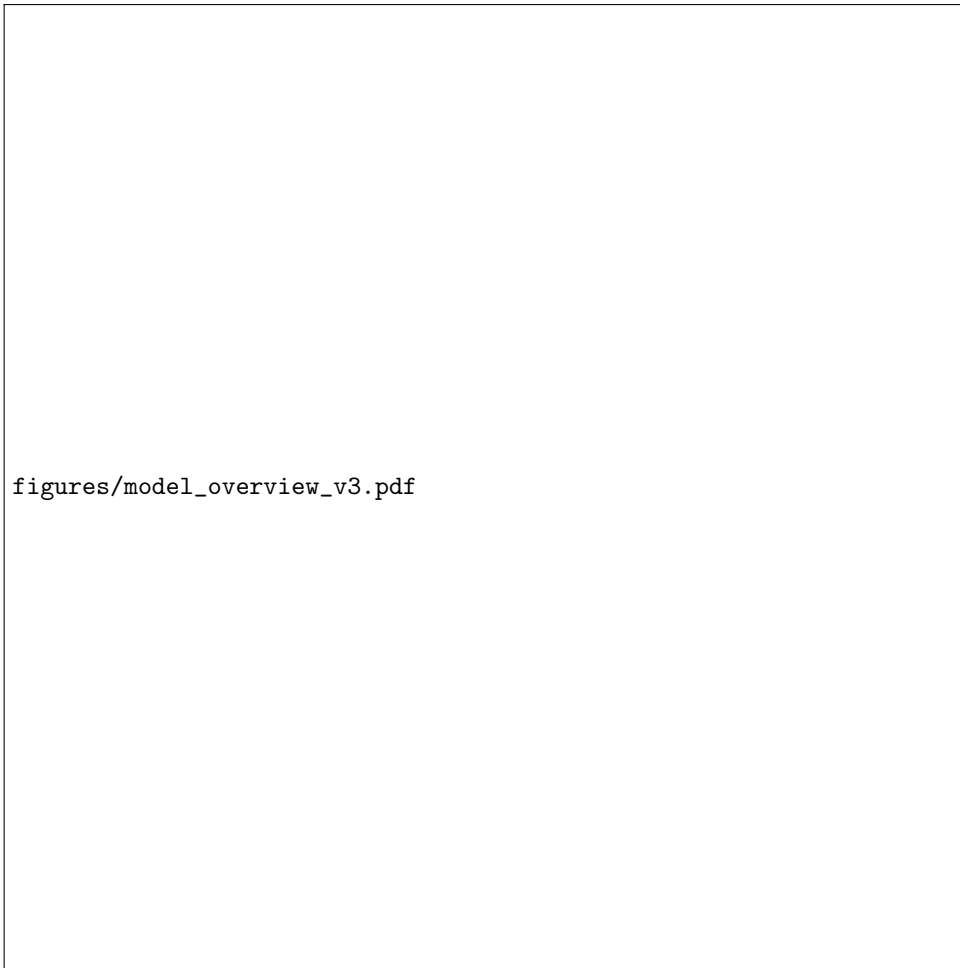


Figure 1: Overview of the FS-SINR **Lange2025Feb**

- Test: multi-label plots → zero shot object detection

Participant: Luciano Dourado

- Approach: filter out background using attention based segmentation using prototype guidance
- Train narrow ViT to match baseline classifier (DinoV2) classification matrix and calculate attention map to find relevant regions
- Use DinoV2 to classify region patches, use grid assembly to search around patch

2.3 Promises and pitfalls of foundation models for the natural world, Lauren Gillespie (MIT)

- Rapid change to environment
- Requires new models: foundation models
- CRISP incorporates multimodal unlabeled data, improves performance across many species detection and range labels (esp. low observation species)

2.3.1 AnimalCLEF Overview

- Challenge: identify *individuals* (e.g. a very specific turtle) given a database of known individuals
- Also: unseen individuals, unclear images, non-overlapping
- Challenges
 - Individual is present, ...

2.3.2 BirdCLEF Overview

- Bioacoustic surveys: use as restoration markers
- Goals:
 - identify taxonomic groups
 - experiment with limited training data
 - experiment with unlabeled data

2.3.3 FungiCLEF Overview

- Few-Shot ID with few samples
- Data: photos, description, metadata, satellite, climate
- Public leaderboard very different from private, takeaway: be robust!
- Different ensemble types, etc.
- Vision-Only Pipelines, Contrastive learning and prototypes helpful!

Participant: Anthony Miyaguchi, GATECH@LifeCLEF

- DS@Georgia Tech - big Data Science group, a lot of publications!!
- PlantCLEF approach: embeddings in kNN setting, adding GEO-info and Priors help a bit
- FungiCLEF: vLLM bad with just prompting, better: interpolating embedding subspaces
- BirdCLEF: Best Working notes, Tokenize Audio dataset (spectrogram), then train on dataset using word2vec+skip-grams, build linear model on top - very efficient, good for deployment!

3 Wednesday

3.1 AI Evaluation Should Make AI Predictable

- Rate LMs by capabilities (as new metrics)
- Taxonomy of LM problems - apply to benchmarks, LM benches measure different things that they claim to (i.e. math tests lang understanding) (Q: the taxonomy is annotated using GPT, isn't this a weakness?)
- Enables the plotting of levels as Spidercharts

3.2 Conference Sessions II

3.2.1 SimpleText Best of Labs in CLEF-2024: Application of Large Language Models for Scientific Text Simplification

3.2.2 Simplified Longitudinal Retrieval Experiments: A Case Study on Query Rewriting and Document Boosting

- Longitudinal evaluation: they provide datasets that can be evaluated for over longer timespan using containers etc.
- Snapshots of datasets
-

3.2.3 Better Call Claude: Can LLMs Detect Changes of Writing Style?

- Identify sentence boundaries
- Goals: benchmark 0-shot on sentence lvl, baselines comparisons, semantic similarity vs. stylistic cues
- Claude has good 0-shot performance, semantic similarity correlated to stylistic changes (?)

3.3 Conference Sessions III + More Labs intro

3.3.1 From Uniform to Unique: Adaptive K-12 Assessment Using Large Language Models

- Generate and assess questions from Kindergarten to 12th Grade
- Use Bloom's taxonomy to instruct model (Remember, Apply, Evaluate) and generate MCQ
- Suppress guessing

3.3.2 Lab Introductions

PAN@CLEF

- AI author attribution¹
 - Binary classification: AI generation? (with Builder/Breaker (red/blue teams), similar to NLP class of Roman Kern) - text with obfuscation; baseline: binoculars, TF-IDF
 - Classify extent of AI gen
- Multilingual detoxification: classification, de-toxify based on keywords; some varied baselines
- Multi-Author Style change detection
- Generative Plagiarism detection

EXIST@CLEF

- Focuses on Benevolent Sexism (e.g. underlying, cultural stereotypes)
- Human Annotations: very varied annotations, embraced as different opinions - target: soft classification
- Novelty: tiktok videos!
- 300k annotations, bias attention
- Sexism classification (binary), direct/ reported/judgemental, kind of sexism (multilabel!), multilingual, multimodal

SimpleText

- Sentence & Document level simplification
- Measure hallucinations in sentence outputs from last years

QuantumCLEF

- Eval QC algorithms
- Foster understanding & build community for QC+IR
- quantum annealing: setup qbits and search for energy minimum
- QUBO: quadratic and binary optimisation – set for IR with retrieval metrics
- Tasks: Feature selection, Instance Selection, Clustering
- Task 1 results: 30x faster, about as effective!

3.4 BioASQ 3/4

3.4.1 MultiClinSum

- Summarize (multiple) long clinical reports
- Multilingual, Semi-Automatically generated summarization
- automatic translation for multilingual tasks
- extractive: only smaller models, bigger models abstractive

3.4.2 BioNNE-L

- Nested Entity Linking
- Multilingual challenge, terms missing in some languages – difficult to reconstruct (Russian, ...)
- Shared dictionary
- Ambiguous terms, UMLS coverage limited – joint dictionary with Russian
- Approach: BERGAMOT - BERT+Graph Encoder and bring together in space to align dictionaries

¹https://bladerunner.fandom.com/wiki/Voight-Kampff_test

3.4.3 ElCardio - Clinical Cardiovascular diseases

- Task: coding (ICD-10 system) for multilingual setting, lack in low-resource languages of discharge letters & extracting code mentions
- similar to gutbrainIE → link entities to ICD-10
- identify all ICD-10 mentions within doc (reverse process)

4 Thursday

4.1 Do we co-evolve with what we design? DevOps, AGI, and Human Fragilities

- Thoughts about how we co-evolve with AI, bio-inspired
- How does exponential growth affect/interact, or is it sigmoid? - how will this affect policy, how to move to stable society away from exp. growth

4.2 Main Conference Session III

4.2.1 MedAID-ML: A Multilingual Dataset of Biomedical Texts for Detecting AI-Generated

- Fake medical literature detection!
- AI generated text generation for multilingual detection

4.2.2 Selective Search as a First-Stage Retriever

- Make search more efficient, distribute web indices (effectively), sparse search...
- Distribute documents by clusters in distributed search
- Approach: use this only as first-stage retrieval, but only care about first documents (i.e. 1000)
- Rank biased Recall (how does this differ from nDCG@k)
- Central problem: which shard (cluster) to take - different approaches based on vocab, ...
- Problem: some selection algs can make shards 'invisible' - documents may not be retrieved as shard index may not expose or represent them correctly.
- Finding: is possible, but efficiency is still a bit lacking

4.3 Labs Overview III

4.3.1 ImageCLEF

- Since 2003 (!)
- Very multimodality-focused, medical tasks
- Datagen, retrieval, classification
- Tasks:
 - MedicalCLEF: caption, generation, VQA
 - ToPicto: image gen (text+speech to pictogram, mostly finetunes)
 - Multimodal VQA
 - Image Retrieval for Arguments
- a lot of participants, 500 runs (expensive)
- A lot of participants used VLMs, explanations: bbox + heatmaps
- Generation: find closest image from training data for generation

4.3.2 eRISK

- Symptom search for depression and detection
- Rank sentences from reddit to clinical classes, contextualized detection, and conversational detection (earlier detection better!); LM personality detection task (Problem: jailbreaking...)

4.3.3 ELOQUENT

- Voight-Kampff task (AI detection, Blade Runner reference!) as red/blue teams - red team quite good, but none fooled all!
- A lot of misclassified, especially two texts: EU law text + intro to LMs ;)
- Value-Oriented questions, 15 languages - no specific answer; only joint participant report!
- Results: LM have some conservative views regarding live, etc.
- Relevance task: return very concise and relevant output!

4.3.4 CheckTHAT

- Tasks:
 - T1: subjectivity/check whether it should be checked
 - T2: Claim extraction
 - T3: Fact-Checking Numerical Claims
 - T4: Scientific Web Discourse: check and identify mentions

4.3.5 TalentCLEF

- Human Capital Management (??)
- HR: very digital, job portals...
- Tasks: Job Title Matching; Skill Prediction from Job Titles

4.4 ImageCLEF

4.4.1 Training Data Analysis and Fingerprint detection

- Synthetic data generation important for medicine (privacy)
- Problem: generative methods have fingerprints in them...
- Task 1: determine which images were used in training, results poor – interesting divide between tasks, reason not fully clear
- Task 2: link to sets of datasets, very high results??

4.4.2 Medical Concept Detection + Captioning

- Concept detection from images (img2text), evaluation using brevity and correctness
- Then explain with bbox, evaluated using radiologist professional (no formal eval, Likert-Scale) – i.e. GradCAM / IG
- Maybe next year as task? very interesting!!

4.4.3 Visual Question Answering and Synthetic Image Generation for Gastrointestinal Tract

- VQA: what, where, how many (polyps) in image- evaluated using BLEU
- Synthetic Data generation based on prompt

4.4.4 Visual Question Answering: Dermatological VQA

- Task 1: Segmentation Maps, solutions mostly finetuned domain models
- Task 2: 'predefined' questions from ontology

4.4.5 ImageCLEFtoPicto

- AAC: augmentative and alternative communication
- Very focused on pictograms, represents ideas & notions
- Currently: a lack of training, and very expensive (+awareness)
- Task: French Text/Speech 2 pictogram
- Very few participants, french-only

4.4.6 Multimodal Reasoning

- Many VQA: very simple questions, images loosely linked to text
- Their benchmark: multilingual (13 languages), multiple-choice, difficulty levels
- Task: Multiple-Choice Questions from student exams within europe
- Some languages test-only!
- Moderately difficulty, parallel data - exactly the same solution across languages, but big diff in languages (e.g. serbian - Cyrillic alphabet!)
- Everyone used VLMs (Qwen Vision)
- Future Work: university-level, are models really reasoning?

4.4.7 Image Retrieval/Generation for Arguments

- Illustrate Argument by images
- Evaluated by aspects contained
- Challenge: combine aspects effectively

5 Friday

5.1 ImageCLEF

5.1.1 ImageCLEFmedical

AUEB NLP Group/Archimedes

- Class Assignment: Multiple Vote strategy of CNNs with ResNET (Union, Intersection, ...)
- Captioning: Q-Former with query assignment, InstructBLIP, Cation Gen + medCLIP scoring (retrieval from generated captions)
- Explainability: assignment based on ChatGPT-drawn boxes on

DS4DH Group

- Concept detection: framed as sequence generation, concepts as tokens (hmmm, CUIs have order; a transformer might be correct) & condition on images
- Caption: InstructBLIP, RAG-based on image retrieval, cluster-based on topics

UMUTeam: Fine-Tuning a Vision-Language Model for Medical Image Captioning and SapBERT-Based Reranking for Concept Detection

5.1.2 MultimodalReasoning (Answers of visual highschool questions)

Ayesha Amjad: Visual Question Answering with Structured Data Extraction and Robust Reasoning

- Approach: Image Captioning using gemini + reasoning modeling for answer generation

ContextDrift: Evaluating VLMs' Multimodal, Multilingual and Multidomain Reasoning Capabilities via Thinking Budget Variations and Textual Augmentation

- Similar Approach, but visual model and prompt design
- A lot of ablation studies²

MSA: Multilingual Multimodal Reasoning with Ensemble Vision-Language Models

- OCR + vLLM
- + Ensembling

²https://www.dei.unipd.it/~faggioli/temp/clef2025/paper_194.pdf

5.1.3 MEDIQA-MAGIC

DS@GT

- Emulate Collaborative Reasoning of Physicians
- 7 vLLM + orchestrators, combination of reasoning ...

IReL, IIT(BHU): Tackling Multimodal Dermatology with CLIPSeg-Based Segmentation and BERT-Swin Question Answering

5.1.4 MEDVQA

Gaurav Parajuli (JKU, Linz): Querying GI Endoscopy

- LoRA finetuned vLLM

Sujata Gaihre

- Similar approach

Krishna Tewari

- Data Augmentation/Preprocessing!

5.2 Closing Ceremony

- New CLEF challenges ;)

6 Posters